

Introduction

What is Linux



After finishing this course you will gain:

- Installing Linux in different ways
- Review of the Linux Graphic Interface
- Easily work with Linux Terminal
- Working with Files, Directories, file contents
- Be familiar with linux file tree
- Working with filters
- Setting up servers: NTP, DNS, DHCP, Nginx , Apache, MySQL, PostgreSQL, NFS, Mail Servers
- Virtualization technologies: Proxmox, Vmware, VirtualBox, ESXi
- Firewalls: Opensense, Pfsense
- Monitoring systems: Zabbix, Prometheus , Grafana

Statistics

- Maybe you are thinking – **Windows claims 82.56% of the world's OS market.**
- In January 2021, 1.93% of all desktop operating systems worldwide ran on Linux.

Mind-blowing Linux Statistics

- In 2021, 100% of the world's top 500 supercomputers run on Linux.
- Out of the top 25 websites in the world, only 2 aren't using Linux: <https://www.alexacom/topsites>
- 96.3% of the world's top 1 million servers run on Linux.
- 90% of all cloud infrastructure operates on Linux and practically all the best cloud hosts use it.
- 85% of all smartphones are based on Linux.

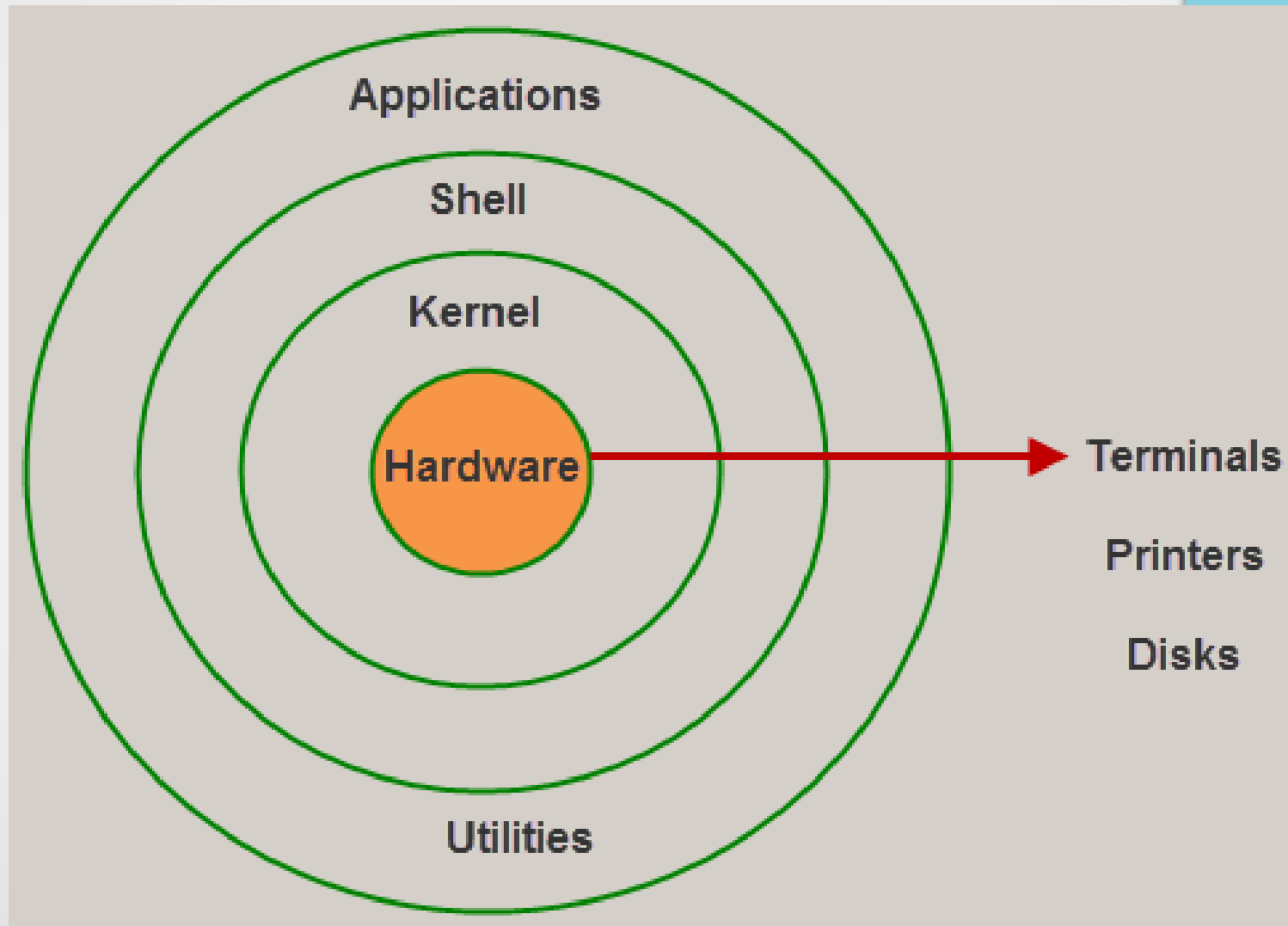
Linux Usage Statistics, 2021

- 25.3% of professional developers used Linux in 2018.
- 36.7% of the websites with known operating systems use Linux.
- 54.1% of professional developers use Linux as a platform in 2019.
- 83.1% of developers say Linux is the platform they prefer to work on.
- As of 2017, more than 15,637 developers from 1,513 companies had contributed to the Linux kernel code since its creation.

Main Terms

- Operating Systems
- Hardware
- Software
- Kernel
- Drivers
- GUI
- CLI
- Shell
- Open Source

Linux Architecture



Linux History

- In 1991 a Finish student named Linus Torvalds created the Linux kernel. Linus first started working on a terminal emulator to connect to Unix systems at his university but over time that project grew into something bigger and eventually became the Linux kernel. Combining the Linux kernel with the GNU Foundation's command line tools created a complete Unix-like operating system. Shortly after the release of the Linux kernel collections of various software combined with the Linux kernel started to appear. These sets of software became known as Linux distributions.

Linux vs Windows

- Usage in PC
 - Compatibility
 - Gaming
 - Easy to use
- Security
 - Privacy
 - Open Source
 - Licence
 - Reliability
 - Multi-choice (distro)
 - Programming
 - Task automation
 - Customization
 - Hardware Support
 - System Performance
 - Meaningful Error messages(Logs)
 - Package managers (apt, yum)
 - Powerful shell
 - Portability

Big companies powered by Linux

Companies That Use Linux



WIKIPEDIA



Google

TESLA

NETFLIX

IBM

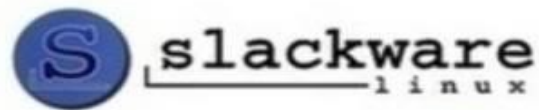


amazon



SPACEX

Linux Distributions



sabayon



yellow dog
linux



fedora



gentoo linux



ubuntu
linux for human beings



xandros
Making Linux Work For You



"An OS for the 21st Century"



open source



opensolaris



debian



archlinux



CentOS



redhat



KNOPPIX



PCLinuxOS



kubuntu



Mandriva